Atmosphere TV

DOOH Venue Incrementality & Media-Mix Measurement

A causal measurement framework, calibrated media-mix model, budget optimizer, and venue revenue + retention models — built to answer the questions Atmosphere's advertising business and its own network economics run on.

How I'd approach this role

Before the demo — the operating thesis behind it: what this role should do for Atmosphere, and how I'd prioritize the first few months as its first data scientist.

Advertising-side measurement

Prove Atmosphere's own incremental foot traffic with methods advertisers can trust — the sell-side differentiator this business runs on. (Answers Q1–Q2.)

Venue-network health

Know which venues are worth more than they're being paid, and which are at risk of leaving — protect and grow the inventory Atmosphere actually sells. (Answers Q3–Q4.)

Not a “quick win vs. foundation” tradeoff — do both in one move

A stratified RCT geo-holdout is the fastest trustworthy result available — and it doubles as the calibration anchor every other method here relies on: synthetic control's placebo check, the MMM's scale, and the venue models' causal-value feature. Ship it first and the “quick win” and the “foundation” are the same deliverable. The venue-network data pipeline (economics + retention features) doesn't have to wait on it — it builds in parallel.

What “done” means

Sales and account teams making different decisions with it: a trusted lift number in a client pitch, a budget conversation anchored to a real response curve, an outreach list ranked by value × risk instead of instinct — not just a model with a good accuracy number.

What follows is one concrete example of applying this approach — a synthetic demo, validated end to end against a known ground truth.

Honest scope

Before the demo that follows: the scope caveats specific to this project, stated up front rather than saved for the closing slide.

Synthetic data	Every figure in this project is synthetic, with a known injected ground-truth effect used specifically to validate that each method recovers it before trusting it conceptually. Not a claim about any real company's data.
Media-effectiveness & rate-card figures	Dwell time, screen count, ad rate card, and similar parameters shown are illustrative demo values, not researched real industry benchmarks. In production: Nielsen OOH, DSP data (e.g. Vistar), or Atmosphere's own play logs and rate card.
Multi-touch attribution — deliberately not built	Atmosphere's ambient-screen model has no individual-level, cross-venue touchpoint log by default. Building MTA would require purchased mobile location/device-matching data — a real but non-default assumption, so it's scoped out rather than forced.
Online-purchase attribution — also out of scope	Every method here measures incremental foot traffic, not downstream online purchases. Tying exposure to e-commerce conversions needs a device-matched exposure-to-transaction panel — a further non-default assumption layered on top of MTA's.
Cost & revenue assumptions	The $/frequency-unit figures behind the budget allocator, and the ad-rate-card behind the venue-revenue model, are illustrative, editable placeholders — in production these come from Atmosphere's own rate card by venue type and daypart.
Retention model — its own caveats	Churn-hazard, competitive-geo, and engagement-signal parameters are illustrative, not researched attrition benchmarks. geo_cluster is deliberately excluded (small-sample tradeoff); realized_ad_revenue's importance is a venue_type confound, not a causal driver; no prospect-side retention model.

Four questions this project answers

Q1–Q2 answer the advertising side (what Atmosphere sells); Q3–Q4 answer the venue-network side (what Atmosphere runs).

1

Did the campaign actually work?

Isolate the TRUE incremental foot traffic caused by ad exposure — net of seasonality, trend, and each venue's own baseline pattern.

The sell-side differentiator: measurement Atmosphere's go-to-market team can take to market and clients can trust.

2

How should budget be spent?

Given a fixed weekly budget, how should it split across restaurants, gyms, bars, and waiting rooms — accounting for each venue type's own diminishing-returns curve?

The media-planning / pricing question advertisers ask before they commit spend.

3

Where's the revenue upside?

Which existing venues are under-monetized relative to their own traffic and quality — and which prospective venues are worth prioritizing for expansion?

Atmosphere's own buy-side value question.

4

Which venues are at risk?

Which existing venues are likely to leave the network, and which ops-visible signals predict it — combined with Q3's revenue into one value-×-risk priority.

Atmosphere's own buy-side risk question.

Design: one pipeline, confidence flows from experiment to product

1 — Q1

RCT geo-holdout

High confidence

Randomized treated/holdout venues. Balance directly verified.

2 — Q1

Synthetic control

Moderate confidence

Covers historical, non-randomized campaigns. Validated with placebo tests.

3 — Q2

MMM, RCT-calibrated

Shape + scale

Adstock/saturation shape from aggregate data; scale pinned by the RCT.

4 — Q2

Budget allocator

Exact DP

Multiple-choice knapsack over calibrated response curves.

5 — Q3

Venue revenue model

GBT, honest OOF

Venue characteristics + the calibrated lift (steps 1–3) as a feature. Flags under-monetized venues; ranks expansion prospects.

6 — Q4

Venue retention model

GBT classifier, honest OOF

Ops-visible signals (engagement, uptime, complaints, outreach) + realized revenue. Flags at-risk venues; combines with Q3 into one value×risk priority.

Answers Q1 — did the campaign work?

Answers Q2 — how should budget be spent?

Answers Q3 — revenue upside?

Answers Q4 — retention risk?

Steps 1–2 answer Q1: RCT for randomized campaigns, synthetic control extending coverage to historical, non-randomized ones. Steps 3–4 turn that into Q2: MMM shapes the response curve, the RCT pins its scale, and the allocator solves the exact split. Step 5 answers Q3: the venue-revenue model reuses the calibrated per-exposure lift from steps 1–3 as a feature. Step 6 answers Q4: a separate churn classifier flags at-risk venues, then combines its risk score with step 5's revenue into one value-×-risk priority — one connected system feeding all four answers, not four separate projects bolted together.

Synthetic data with a known, injected ground truth

4 venue types × 100 venues each, 104 weeks (52 pre-period + 52 campaign)

RCT pool

Randomly split treated / holdout within venue_type × geo_cluster × traffic-tier strata for a fixed benchmark campaign. This is a real Atmosphere-designed measurement product, not a historical campaign.

Used to anchor confidence — assignment is verified, not assumed.

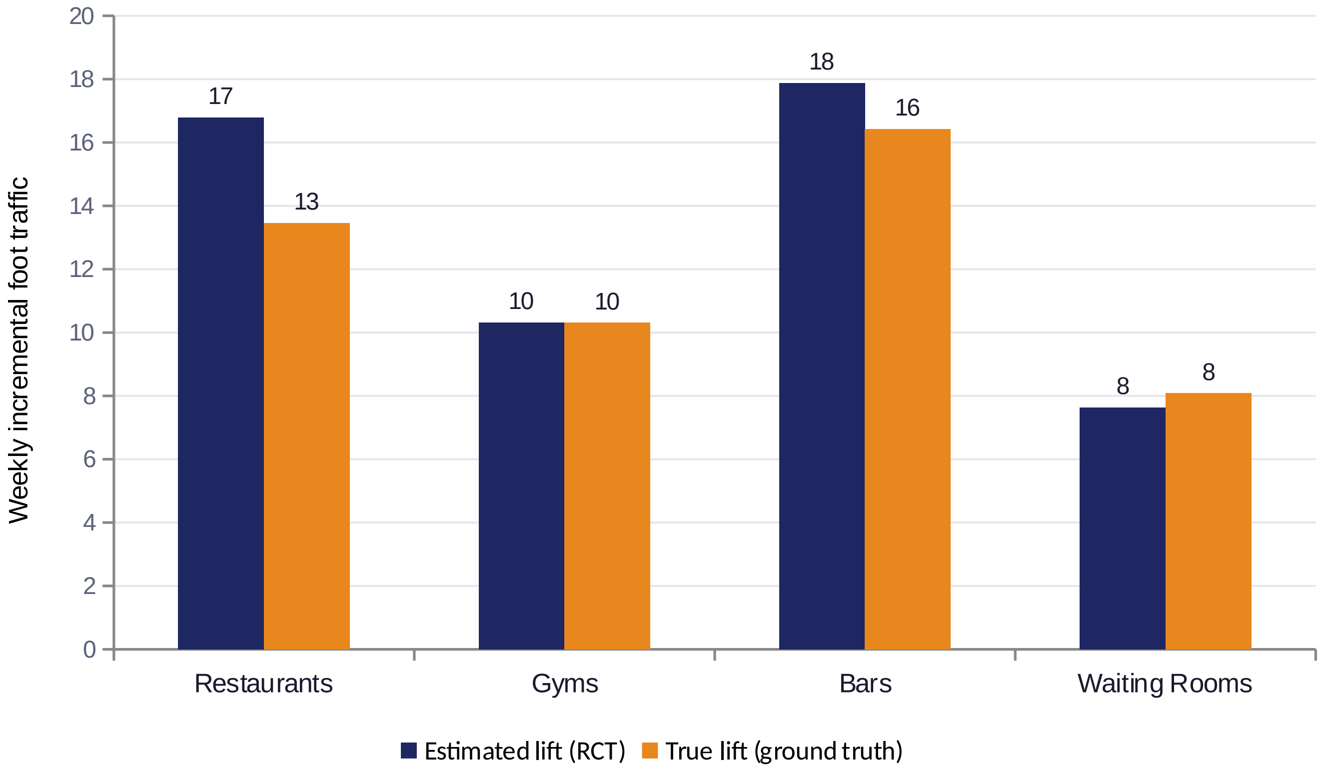
Observational pool

Advertisers activate venues themselves — non-randomly. Higher-baseline-traffic venues are systematically more likely to be activated (a real selection mechanism), and weekly frequency varies venue-to-venue.

A naive before/after comparison here is confounded by selection — this is what synthetic control and the MMM have to work around.

RCT geo-holdout — high confidence

2/16 balance tests sig.

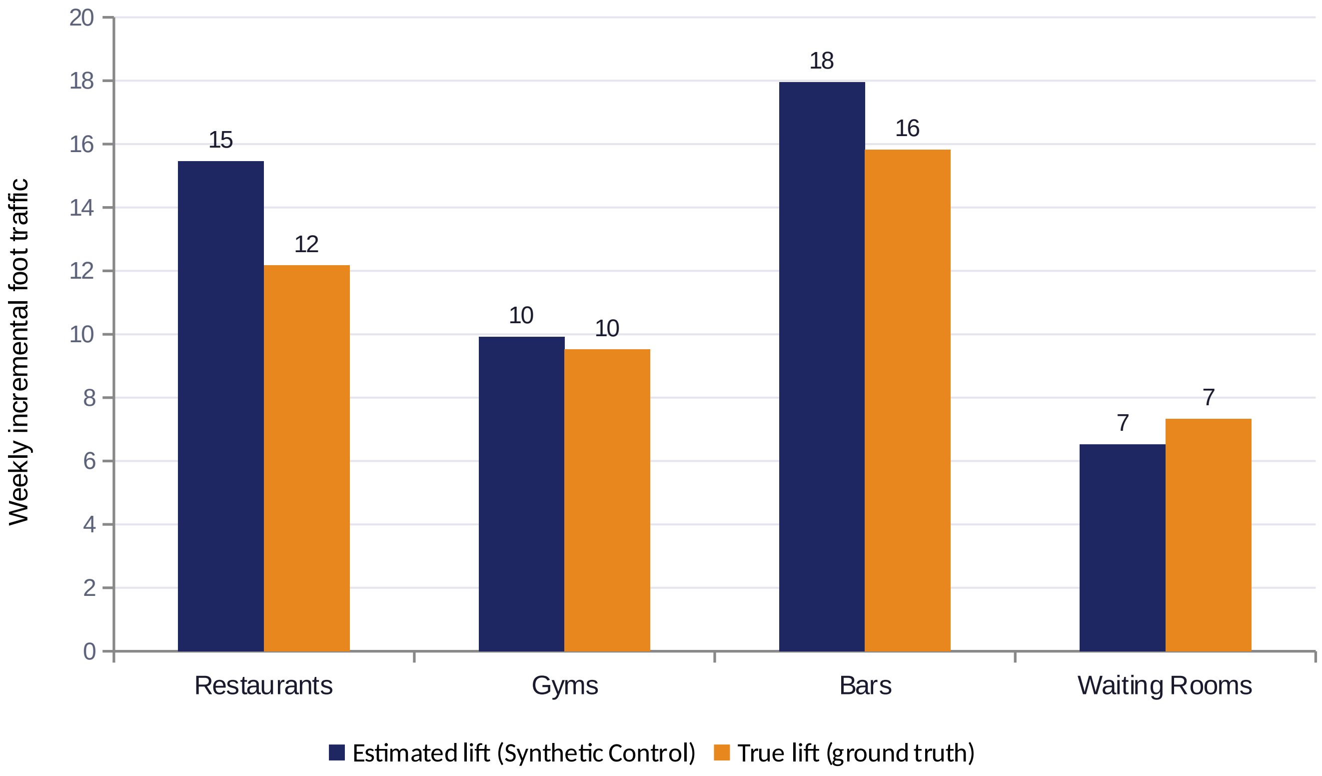


Venue type	95% CI	p-value
Restaurants	[11.1, 22.4]	< 0.001
Gyms	[5.2, 15.4]	< 0.001
Bars	[13.3, 22.5]	< 0.001
Waiting Rooms	[5.1, 10.2]	< 0.001

Pre-treatment balance check: only 2 of 16 covariate × venue-type tests were significant at p≤0.05 — in line with what chance alone predicts. Randomization worked as designed, not assumed.

Synthetic control — moderate confidence

Covers historical, non-randomized campaigns — the identifying assumption can't be directly tested the way randomization can, only checked indirectly.

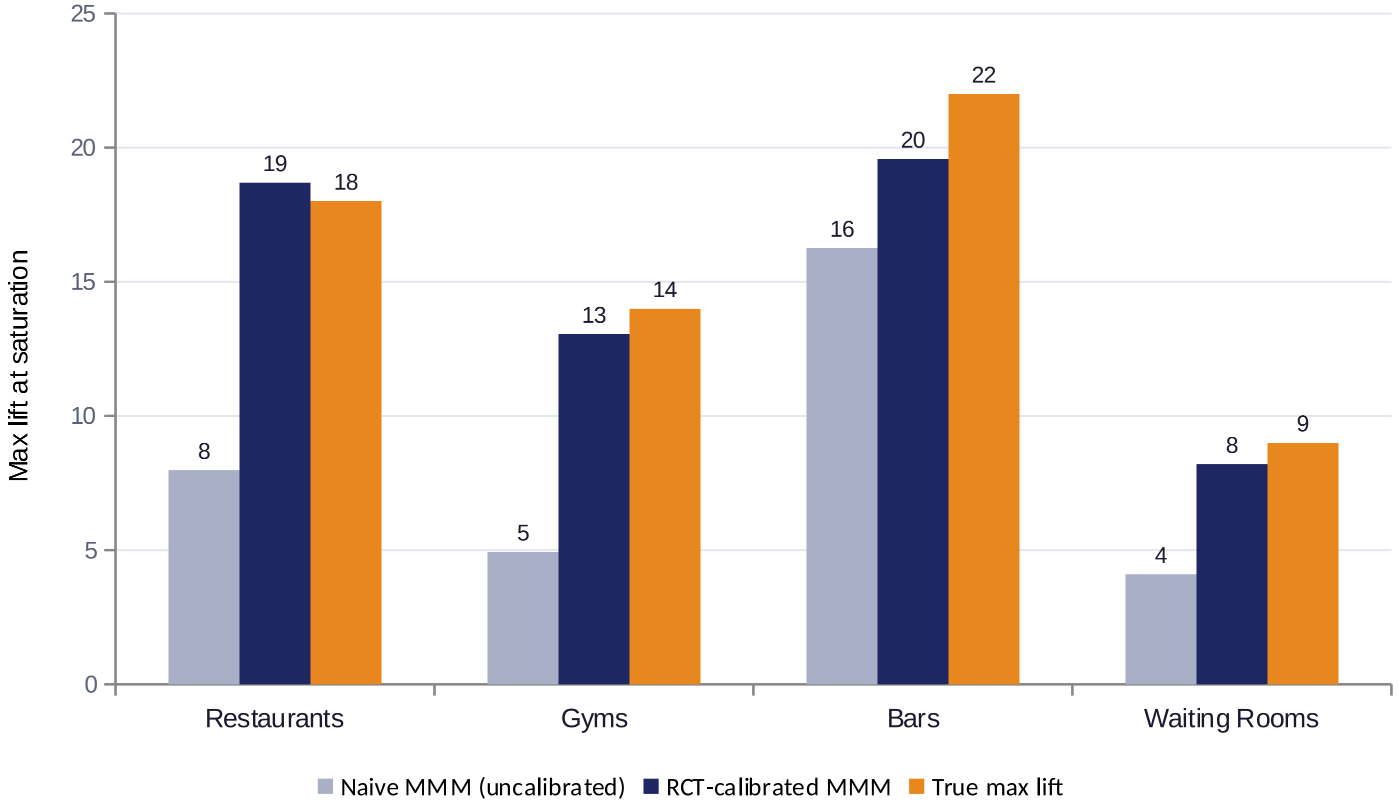


Venue type	Placebo p	Pre-RMSPE
Restaurants	0.160	26.4
Gyms	0.050	14.4
Bars	0.100	17.8
Waiting Rooms	0.110	8.3

Two validation checks, since synthetic control has no closed-form standard error: pre-period fit quality (RMSPE — poor fits are flagged and excluded from the aggregate estimate) and an in-space placebo test on every donor venue.

MMM: an uncalibrated model would have underpriced results

Adstock + saturation shape comes from the aggregate weekly series; scale is pinned by the RCT's high-confidence estimate.



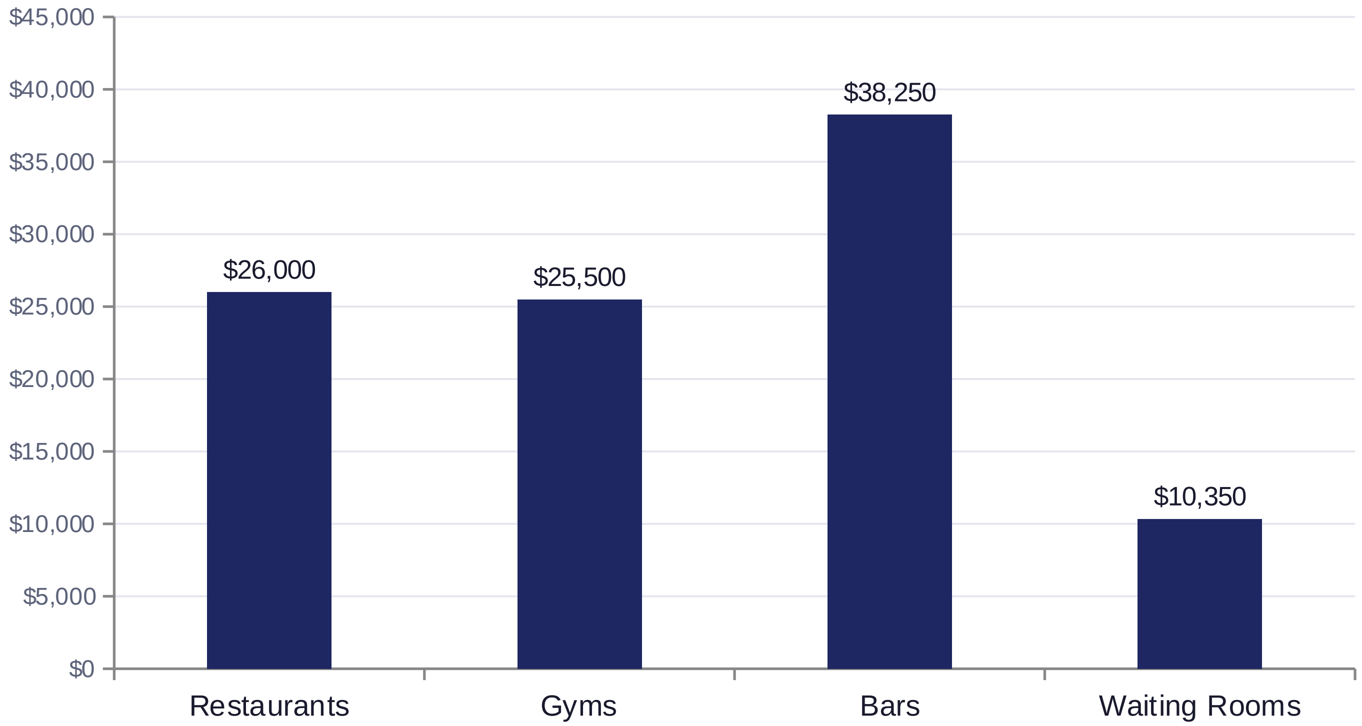
Calibration adjustment

Restaurants	+134%
Gyms	+165%
Bars	+20%
Waiting Rooms	+100%

For 3 of 4 venue types, the naive observational fit understated true incremental lift by 100–165% — the kind of gap real MMM practice cites as its core identification problem, and exactly why the RCT anchor matters.

Budget allocator: exact DP, not a greedy walk

A Hill/S-shaped response curve is CONVEX before its inflection point — a greedy "spend the next dollar on whichever venue type currently looks best" heuristic isn't guaranteed optimal there. An earlier greedy version of this allocator actually underperformed a naive equal-split baseline; the DP formulation below has no concavity requirement and is guaranteed to find the grid-optimal allocation.



$100,000 weekly budget example

$20,000 budget **+48.6%**

vs. naive equal-split

$50,000 budget **+11.1%**

vs. naive equal-split

$100,000 budget **+2.4%**

vs. naive equal-split

The optimizer's edge is largest when budget is scarce — exactly when allocation decisions matter most.

The other side of the business

One system answers the advertising side. The same system answers the venue-network side too.

Advertising incrementality

- Prove and price advertising incrementality
- Feeds go-to-market: sell-side differentiator, client trust
- RCT + synthetic control + calibrated MMM + budget DP

Feeds the venue-economics model as a validated causal-value input (calibrated per-exposure lift), not a separate silo.

Venue network economics

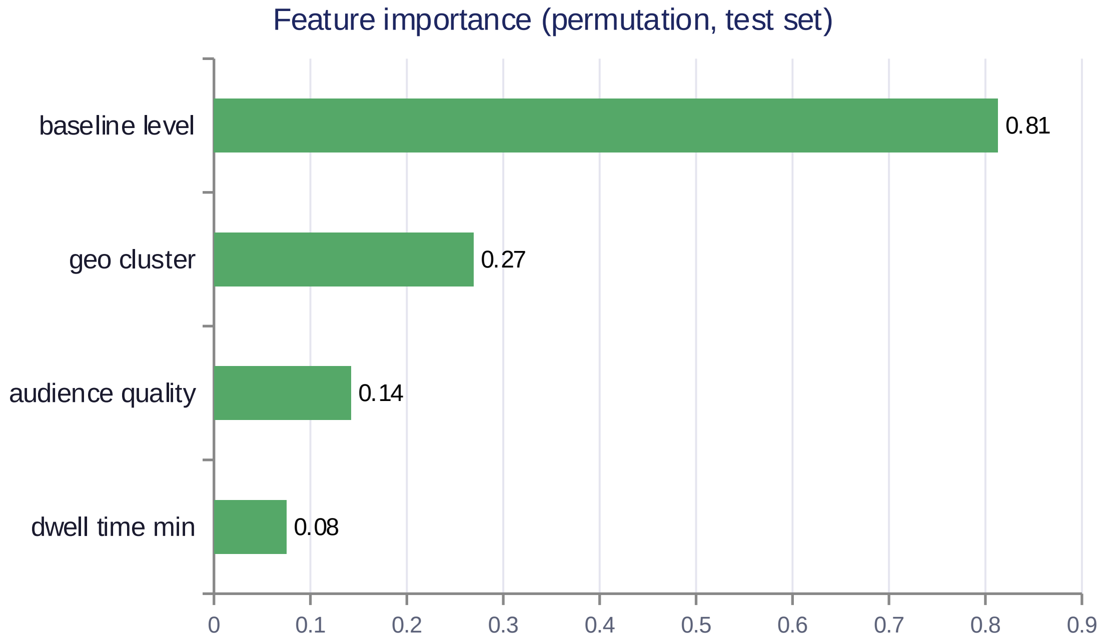
- Predict realized ad-revenue (Q3) and 90-day churn risk (Q4) from characteristics and ops signals
- Flag under-monetized (Q3) and at-risk (Q4) existing venues for sales/ops follow-up
- Rank prospective venues for expansion priority (Q3)
- Combine Q3 revenue + Q4 risk into one value-×-risk priority quadrant
- GBT regressor + classifier, honest out-of-fold evaluation throughout

"Smarter on both sides of the business" — the venue-side model shares its data foundation and causal-value input with the advertising side, rather than being a disconnected second project. Results on the next slide.

Predicting venue ad-revenue, honestly

Gradient-boosted trees on observable venue characteristics + the calibrated per-exposure lift as a feature.

0.84	22.1%	0.94	90% vs 28%
Held-out R^2	Held-out MAPE	Corr. w/ latent true potential	Flagged-tail latent gap rate



Top under-monetized venues (of 20 flagged)

Venue #85 (Restaurants)	-62%
Venue #309 (Waiting Rooms)	-61%
Venue #53 (Restaurants)	-61%
Venue #247 (Bars)	-60%

Top prospect venues (expansion priority)

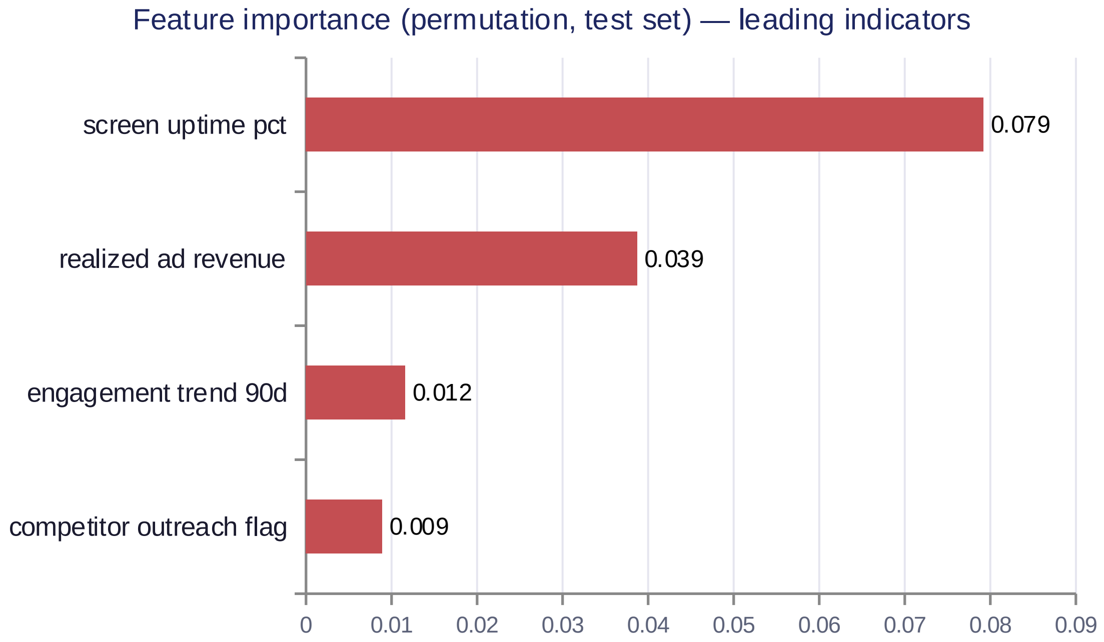
P0018 — Bars (existing market)	$684
P0055 — Restaurants (existing market)	$524
P0000 — Bars (new market)	$500
P0001 — Bars (new market)	$496

Flags come from 5-fold out-of-fold predictions, never a model scoring the venue it was trained on. The calibrated-lift feature carries near-zero importance — honest, since it's constant within venue_type once venue_type itself is a feature.

Predicting venue churn risk, honestly

Gradient-boosted classifier on ops-visible signals (engagement, uptime, complaints, competitor outreach) — the other half of "what makes a venue valuable and what puts it at risk."

0.65	**0.27**	**0.73**	**100% vs 31%**
Held-out AUC (oracle ceiling: 0.73)	Held-out PR-AUC	OOF corr. w/ latent true risk	Flagged-tail latent risk rate



Top at-risk venues (of 20 flagged, by OOF risk)

Venue #281 (Bars) — 60%
engagement declining → content/placement review

Venue #200 (Bars) — 55%
engagement declining → content/placement review

Venue #12 (Restaurants) — 52%
uptime issue → dispatch technical support

Priority quadrant (Q3 revenue × Q4 risk)

Save now (high value, high risk)	136
Monitor (low value, low risk)	136
Protect (high value, low risk)	64
Low priority (low value, high risk)	64

Flags come from 5-fold OOF predictions; AUC/PR-AUC read modest next to the revenue model's R^2 only because the oracle ceiling here is 0.73 AUC. realized_ad_revenue's importance is a venue_type confound, not a causal driver. Each "Save now" / flagged venue also carries a SHAP top driver (from the fold model that never saw it) mapped to a concrete action — technical dispatch, AM retention call, content review — not a generic "reach out."

Takeaways

1. Match the method to the assignment mechanism — RCT where randomization is designed, synthetic control where it isn't, each with its own validation check.

2. An uncalibrated MMM understated true incremental value by up to 165% here — experiment calibration isn't optional polish, it's the identification fix.

3. Optimization needs the right algorithm for the curve's shape — a greedy heuristic silently lost to a naive baseline on non-concave response curves; the exact DP formulation doesn't.

4. One connected system, not four separate projects: the venue-revenue model reuses the causal pipeline's per-exposure value as an input; the retention model then combines its own churn-risk score with that revenue prediction into one value-×-risk priority quadrant — both models recover their respective latent ground truths (0.95 and 0.73 correlation) before either is trusted.

Thank you